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Lightweight ACG-YOLO with Collaborative Attention Mechanisms for Real-Time Public Safety Surveillance and Edge Deployment

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Abstract

Real-time detection of violent acts and hazardous weapons in edge-deployed public safety surveillance remains challenging because resource-constrained devices must balance detection accuracy and computational efficiency, particularly for small-sized weapon targets in complex multi-person interaction scenes. To address these issues, this paper proposes ACG-YOLO, a lightweight collaborative detection framework built upon the YOLOv11n baseline. The model adopts a three-level design involving efficient feature downsampling, spatial localization enhancement, and global context modeling. Specifically, an ADown module reduces information loss during downsampling through a dual-branch structure, helping preserve fine-grained features of small targets; a Coordinate Attention mechanism enhances spatial localization by encoding horizontal and vertical positional information; and a Global Context Enhancement Module (GCEM) incorporates lightweight global contextual information to improve feature representation in complex violent interaction scenes. Experimental results on a violence and weapon detection dataset demonstrate that ACG-YOLO achieves an mAP@0.5 of 89.2%, representing an improvement of 2.9 percentage points over the baseline YOLOv11n, while reducing the parameter count by 19.8% and computational cost by 19.0%, with only 2.07 M parameters and 5.1 GFLOPs. Ablation experiments further examine the contributions of the individual components and their combinations. This research provides a practical solution for real-time early warning and emergency response in resource-constrained public safety surveillance scenarios.

Keywords: lightweight object detection; YOLOv11n; attention mechanism; edge deployment; public safety surveillance; small-target detection

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Received: 7 August 2026
Revised: 12 September 2026
Accepted: 15 September 2026
Published: 17 September 2026

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1. Introduction

Intelligent video surveillance has become the core perception layer of "safe cities" and intelligent social governance systems, and is widely deployed in public transportation hubs, campuses, commercial complexes, and border checkpoints, serving as indispensable infrastructure for maintaining public security [1,2]. In such scenarios, real-time detection of violent acts and hazardous weapons is of particularly urgent practical significance. Such incidents are often sudden, rapidly evolving, and potentially hazardous. Timely and accurate detection can provide an early warning for emergency response and help mitigate the consequences of such incidents [3]. However, current public safety surveillance

still relies heavily on manual monitoring, whose inherent shortcomings are becoming increasingly evident: prolonged monitoring leads to human visual fatigue and a marked increase in missed detection rates; human response delays under multi-stream concurrent video feeds often cause missed opportunities for timely intervention; and the sheer volume of real-time video data far exceeds the upper limit of human processing capacity. These limitations motivate the development of automated visual detection methods suitable for edge deployment to assist human operators in timely threat identification and response.

However, directly transferring generic object detection algorithms to violence and weapon detection tasks faces three deeply coupled technical challenges: Feature degradation of small targets—handheld weapons occupy only dozens to hundreds of pixels in surveillance images; after multi-layer convolutional downsampling in the backbone network, high-frequency details such as edges and textures are easily overwhelmed by background noise, resulting in a severe attenuation of feature saliency [4]. Difficulty in modeling complex spatial interactions and long-range dependencies—violent conflicts involve multi-person limb entanglement, abrupt relative position changes, and dynamic occlusions, lacking fixed geometric constraints, with requirements for global contextual semantic understanding that far exceed those of conventional detection tasks [5]. Stringent computational resources on edge devices—front-end surveillance devices are mostly low-power embedded platforms with extremely limited compute and storage capabilities; high-accuracy large models struggle to meet real-time latency constraints, while aggressive lightweight design tends to significantly impair small-target perception accuracy [6].

More critically, these three factors form a Pareto-type constraint: increasing resolution or network depth may improve small-target perception, but substantially increases computational cost; model compression can improve inference efficiency, but excessive compression may reduce feature discriminability. Therefore, seeking an optimal trade-off among detection accuracy, inference efficiency, and edge feasibility has become a core scientific challenge that urgently needs to be addressed in this field.

To address the above challenges, this paper proposes a lightweight violence and weapon detection framework for public safety surveillance scenarios, termed ACG-YOLO (Adown + CoordAtt + GCEM–YOLO). Built upon YOLOv11n, it achieves a balance between detection accuracy and deployment efficiency via the synergistic optimization of efficient feature extraction, spatial position modeling, and global semantic enhancement. The main contributions of this paper are as follows:

- To address the challenges of small-target weapon detection, complex violent behavior understanding, and real-time deployment on edge devices in public safety surveillance, we construct a lightweight collaborative detection framework, ACG-YOLO. The framework follows a progressive design involving efficient feature extraction, spatial localization enhancement, and global context modeling to balance detection performance and computational efficiency.
- Three complementary components are incorporated to address the above problems: the ADown downsampling module is incorporated to replace conventional strided-convolution operations [7], mitigating detail loss during downsampling through a dual-branch information preservation mechanism while maintaining controlled computational complexity; Coordinate Attention (CoordAtt) is incorporated into multi-scale feature maps to enhance spatial localization by preserving directional positional information [8], improving the model's perception of small weapon targets and object boundaries in complex scenes; and a lightweight Global Context Enhancement Module (GCEM) is designed to supplement global semantic representation capability through low-cost global channel statistics, forming complementary synergy with Co-

ordAtt along both spatial localization and semantic modeling dimensions to enhance understanding of multi-person interactions and complex violent scenes.

- Systematic experiments are conducted on the Violence and Weapon Detection dataset as well as the COCO2017 dataset [9]. The results demonstrate that ACG-YOLO achieves an mAP@0.5 of 89.2%, a 2.9 percentage-point improvement over YOLOv11n, with only 2.07 M parameters and 5.1 GFLOPs, representing reductions of 19.9% and 19.0%, respectively, compared to the baseline. Ablation studies verify the synergistic gains among the constituent modules, and generalization experiments further confirm the model's favorable cross-dataset adaptability.

Furthermore, beyond static surveillance infrastructure, the deployment of lightweight detection models on mobile platforms such as unmanned aerial vehicles (UAVs) introduces additional challenges, where visual perception must operate in tandem with robust flight control and disturbance-rejection mechanisms to maintain reliable performance under environmental perturbations [10,11].

It should be noted that the evaluation of edge deployment feasibility in this work is based on widely accepted theoretical metrics, including model parameter count, computational complexity (GFLOPs), and inference speed (FPS), which have been extensively adopted as effective proxies for edge applicability in prior lightweight detection studies. The training process is conducted on a vGPU platform to ensure reproducibility and fair comparison with baseline methods. The reported complexity metrics, together with the competitive FPS, demonstrate the strong potential of ACG-YOLO for resource-constrained deployment scenarios.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 elaborates on the overall architecture of ACG-YOLO, along with the design and mathematical formulation of each module. Section 4 reports the experimental setup, comparative results, ablation studies, and generalization analysis. Section 5 concludes the paper and discusses future directions.

2. Related Work

2.1. Object Detection Technology in Surveillance Scenarios

Deep learning-based object detection serves as the core perception engine for intelligent video surveillance systems, and its developmental trajectory has profoundly influenced the technical selection for violence and weapon detection tasks. Mainstream detection frameworks can be broadly categorized into two-stage detectors and one-stage detectors. Two-stage detectors, such as Faster R-CNN and Mask R-CNN [12], typically generate region proposals before performing classification and bounding-box regression, which can provide strong detection accuracy but at a higher computational cost. However, they suffer from slow inference speed and high parameter counts, making it difficult to meet the low-latency requirements of real-time surveillance. One-stage detectors, on the other hand, reformulate detection as an end-to-end regression problem, simultaneously outputting bounding boxes and class probabilities in a single forward pass. Represented by SSD [13] and the YOLO series [14], they are widely used in real-time video analysis because of their favorable speed–accuracy trade-off. YOLOv9 introduces programmable gradient information, YOLOv10 optimizes post-processing latency, and YOLOv11 achieves a more favorable trade-off between accuracy and efficiency, establishing itself as a highly competitive baseline model for edge deployment scenarios [14,15].

2.2. Lightweight Object Detection Network

Deploying high-accuracy models on resource-constrained edge devices requires algorithms to achieve a delicate balance between computational efficiency and detection

performance. Howard et al. [16] proposed MobileNet, which employs depthwise separable convolution to factorize standard convolution into channel-wise filtering and pointwise linear combination, reducing both parameter count and computational cost to approximately 1/8 of those of conventional convolution. Sandler et al. [17] introduced the inverted residual structure and linear bottleneck layers in MobileNetV2, effectively mitigating information loss in low-dimensional spaces. Zhang et al. [18] proposed ShuffleNet, which leverages pointwise group convolution and channel shuffling to enable efficient cross-group feature flow. Subsequently, ShuffleNetV2 [19] systematically analyzed the non-equivalent relationship between FLOPs and actual inference speed from the perspective of hardware execution efficiency, further enhancing edge practicality.

In recent years, lightweight detection techniques have continued to evolve. Han et al. [20] proposed GhostNet, which replaces a portion of standard convolutions with cheap linear transformations, compressing model size while maintaining feature richness. Li et al. [21] introduced DSConv, which reduces redundant convolutional computation via variable stride. In terms of lightweight improvements to the YOLO series, Hu et al. [22] proposed Micro-YOLO, which replaces the convolution layer with the largest weight proportion in YOLOv3-tiny with a mobile inverted bottleneck convolution incorporating a squeeze-and-excitation structure. Ge et al. [23] constructed YOLOX-Nano via depthwise convolution and weakened data augmentation, and decoupled the detection head into three branches—classification, background, and regression—to accelerate convergence. Cui et al. [24] proposed LC-YOLO, which employs a Laplacian bottleneck structure to enhance shallow-layer details and achieves adaptive fusion of shallow and deep features via cross-layer attention upsampling. Cai et al. [25] introduced ADown downsampling and C2f structure optimization based on YOLOv11, significantly reducing computational cost while maintaining accuracy. However, the aforementioned works primarily target overall parameter compression and do not specifically address the high-frequency detail fidelity of small-scale weapon targets, resulting in irreversible information degradation of tiny targets at an early stage of feature extraction.

2.3. Attention Mechanisms in Visual Perception

Attention mechanisms, which dynamically recalibrate feature response intensities to enable networks to adaptively focus on task-relevant regions, have become a key technique for improving detection performance. Based on the dimension of operation, they can be categorized into channel attention, spatial attention, and hybrid forms.

In terms of channel attention, Hu et al. [26] proposed the Squeeze-and-Excitation Network (SENet), which compresses spatial information via global average pooling and generates channel weights through a bottleneck fully connected layer, enhancing discriminability with extremely low computational overhead. Wang et al. [27] proposed Efficient Channel Attention (ECA), pointing out that the dimensionality reduction operation in SENet adversely affects channel dependency modeling, and instead adopts one-dimensional convolution to enable local cross-channel interaction, achieving comparable performance with even fewer parameters. In terms of spatial attention, Woo et al. [28] designed a spatial branch in CBAM, which concatenates max-pooling and average-pooling along the channel dimension and then applies convolution to generate a 2D spatial attention map, explicitly highlighting critical regions.

Among hybrid attention mechanisms, the Coordinate Attention (CoordAtt) mechanism proposed by Hou et al. [8] explicitly incorporates positional information into channel attention. It upgrades the conventional channel attention operation—which compresses spatial information into a scalar—into two independent orientation-aware feature maps. Through horizontal and vertical one-dimensional global pooling, it explicitly embeds

positional information into channel attention, while introducing only limited additional computational overhead. This mechanism has been successfully applied to tasks such as safety helmet detection in complex environments. However, the aforementioned attention mechanisms remain confined to local receptive fields or single-scale statistical information, lacking explicit modeling of global long-range dependencies. In violent scenarios involving multi-person limb interactions, semantic associations among distant individuals are crucial for correctly judging behavior, while local attention mechanisms struggle to capture such cross-regional relationships. Although Transformer self-attention [29] is capable of establishing long-range dependencies, its computational cost increases quadratically with the number of spatial positions, which can limit real-time inference on resource-constrained edge devices.

Beyond performance enhancement, recent studies have explored the interpretability of attention-based detection models. Visualization methods such as Grad-CAM have been integrated with YOLO-based frameworks to provide visual explanations of model decisions [30], and human-attention-guided XAI approaches have been developed specifically for object detection [31]. In parallel, hybrid feature extractors combining CNNs with Transformer architectures have emerged to leverage both local detail extraction and global context modeling. For instance, DConvTrans-LKA integrates local and global attention for remote sensing detection [32], and DyFormer combines CNN and Transformer branches to improve small-target detection [33]. These hybrid approaches share a common motivation with our work: effectively fusing multi-scale and multi-receptive-field information.

2.4. Methods for Detecting Violence and Weapons

As a specialized task in intelligent surveillance, violence and weapon detection has developed along two technical routes: video-sequence-based and still-image-based approaches. Video-based methods leverage temporal information to model behavioral dynamics. Sultani et al. [34] formulated violence detection as an anomaly detection problem, leveraging multiple-instance learning to achieve frame-level anomaly localization with only video-level labels. Garcia-Cobo et al. [35] proposed an efficient violence detection method based on human skeleton and change detection, which captures motion patterns via sequences of skeleton keypoints, effectively utilizing behavioral dynamic information while maintaining real-time performance.

Still-image-based methods are compatible with one-stage detectors such as YOLO and are therefore suitable for single-frame real-time detection. Mukto et al. [36] designed an end-to-end real-time crime surveillance system, employing YOLOv5 for weapon detection, MobileNetV2 for violence recognition, and face recognition for identity confirmation, demonstrating excellent performance in real-world scenarios. A unified threat detection model based on YOLOv8 achieves simultaneous recognition of six target categories—non-violence, violence, firearms, knives, sticks, and axes—on over 8500 annotated images. Kumari et al. [37], in their systematic review, comprehensively evaluated the robustness of YOLOv8 through YOLOv11 under challenging conditions such as low illumination, motion blur, and occlusion, providing important experimental benchmarks for algorithm design in complex surveillance scenarios.

In summary, existing methods still suffer from two main limitations. First, most approaches treat violent behavior and weapon detection as relatively independent subtasks in a pipeline manner, without fully exploiting the semantic associations between weapon presence as physical evidence of violence and limb interactions as behavioral context. Second, when dealing with small-scale weapons and complex interactions, the attention modules adopted by existing methods remain primarily focused on local receptive field

recalibration, and have not yet introduced a truly lightweight global context modeling mechanism to capture the long-range dependencies inherent in multi-person interactions.

2.5. Motivation and Problem Formulation

Based on the above literature review, this paper identifies three interrelated and as-yet insufficiently addressed research gaps in the current field of violence and weapon detection:

Gap 1: The trade-off between lightweight design and small-target feature preservation. Existing lightweight methods generally adopt overall parameter compression as their optimization objective, lacking dedicated maintenance of high-frequency detail fidelity in shallow layers. This can attenuate edge and texture information for small-sized weapon targets during early downsampling, limiting the detection performance of lightweight models in surveillance scenarios.

Gap 2: The lack of lightweight modeling capability for long-range dependencies in existing attention mechanisms. Attention computations such as SE, ECA, and CoordAtt are confined to local statistics or directional aggregation within a single feature map, failing to capture cross-regional multi-person interaction semantics. Although Transformer self-attention is capable of establishing global dependencies, its quadratic computational growth precludes real-time inference on edge devices. This limitation restricts the ability of lightweight models to represent global contextual relationships in complex violent scenes.

Gap 3: Functional redundancy and mutual interference when multiple attention mechanisms operate collaboratively. When multiple attention mechanisms are jointly employed within the same framework, insufficient functional separation between spatial and channel attention mechanisms may lead to redundant feature modeling, thereby diminishing their respective independent gains. The existing literature lacks systematic analysis of such collaborative compatibility issues.

The above three gaps are inherently intertwined: Gap 1 provides the foundational quality of feature input for subsequent modules, while Gap 3 imposes collaborative architectural design constraints on the implementation of Gap 2. Based on this progressive design logic—feature preservation, spatial localization, and global context modeling—we integrate ADown, CoordAtt, and GCEM into ACG-YOLO—each addressing one of the above gaps: ADown preserves fine-grained features (addressing Gap 1), CoordAtt enhances positional awareness with extremely low overhead (addressing the spatial localization aspect of Gap 2), GCEM achieves semantic modeling with an approximate global receptive field through lightweight global channel statistics and a bottleneck transformation, while avoiding spatial similarity computation (addressing the global semantic aspect of Gap 2 as well as the efficiency constraints of Gap 3).

Unlike existing lightweight YOLO variants that typically pursue overall parameter compression through backbone replacement or single-attention insertion, ACG-YOLO addresses three deeply coupled challenges through a functional, orthogonal decomposition into ADown (efficiency assurance), CoordAtt (spatial fine localization), and GCEM (global semantic prior). The fundamental distinction lies in the system-level collaborative design: rather than stacking off-the-shelf modules, our framework assigns each module an orthogonal role—ADown preserves fine-grained features during downsampling, CoordAtt provides direction-aware positional encoding, and GCEM supplies global semantic understanding—with their synergistic gains empirically validated in ablation studies and framework-level comparisons.

3. Materials and Methods

3.1. Overall Architecture of ACG-YOLO

We develop ACG-YOLO based on the lightweight YOLOv11n architecture to address small-sized weapon detection, complex interaction modeling, and real-time deployment under resource constraints. The overall architecture is illustrated in Figure 1. ACG-YOLO follows a collaborative design principle of "efficient feature extraction, spatial localization enhancement, and global semantic modeling." Accordingly, ADown, Coordinate Attention (CoordAtt), and the Global Context Enhancement Module (GCEM) are integrated into different stages of the YOLOv11n framework to address complementary aspects of feature representation.

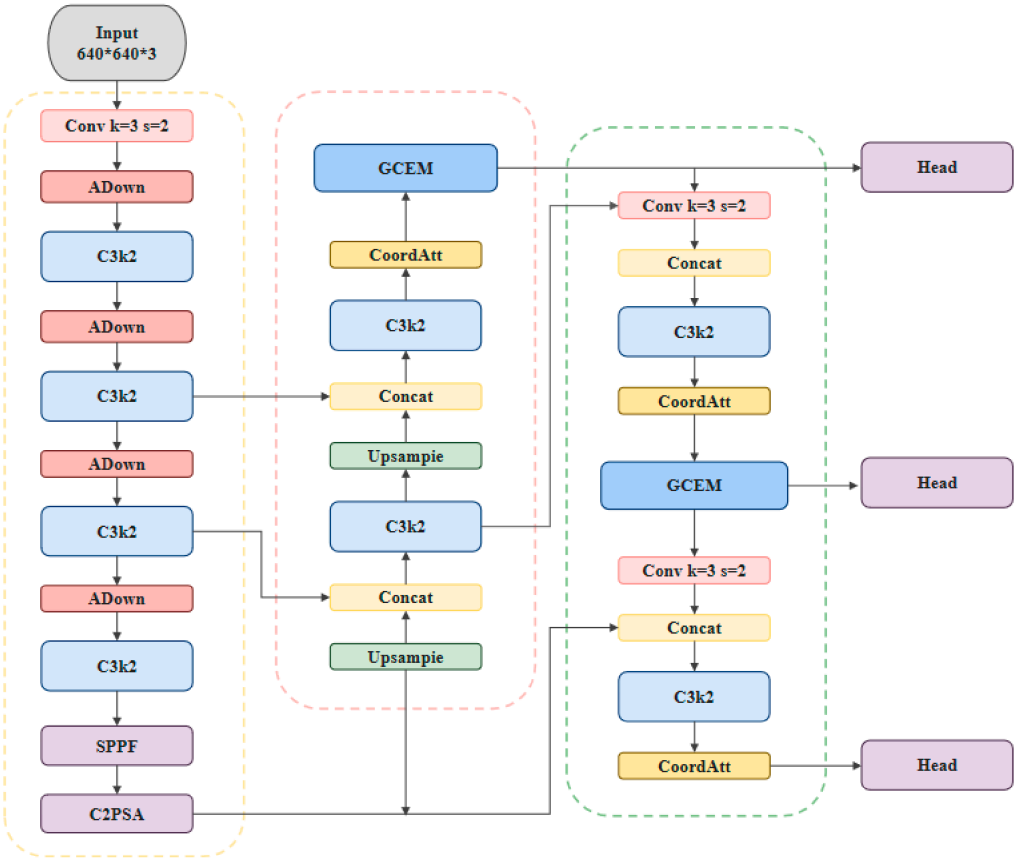


Figure 1. The overall architecture of the ACG-YOLO framework.

Given an input image of 640 × 640 × 3, the backbone first performs hierarchical feature extraction through convolutional and C3k2 blocks. The standard downsampling operations are replaced by ADown at the corresponding stages of the backbone. Through its parallel branches, ADown reduces spatial resolution while retaining complementary local information, thereby alleviating the loss of fine-grained features during progressive downsampling. The resulting hierarchical representations are further processed by C3k2, SPPF, and C2PSA to obtain multi-scale features with increasingly rich semantic information.

The extracted multi-scale features are subsequently delivered to the neck for bidirectional feature fusion. The neck adopts top-down and bottom-up information propagation to combine high-level semantic information with low-level spatial details. Coordinate Attention is incorporated into the feature-processing stages of the neck to enhance spatially discriminative responses while preserving positional information along the horizontal and vertical directions. This spatially aware enhancement is particularly beneficial for small and

partially occluded weapon targets, whose localization depends strongly on fine-grained structural and positional cues.

In addition to spatial localization enhancement, GCEM is integrated into the neck to provide global semantic information. GCEM first aggregates spatial information through global average pooling and then performs lightweight cross-channel interaction through a bottleneck transformation. The resulting channel-wise weights are applied to the original feature representation for semantic recalibration. In contrast to CoordAtt, which emphasizes spatial positional information, GCEM uses global channel statistics to enhance semantically informative features. Thus, the two modules provide complementary enhancement from the spatial and channel-semantic perspectives.

The enhanced multi-scale features are progressively fused within the neck and subsequently transmitted to the decoupled detection heads for classification and bounding-box regression. Overall, the architecture forms a collaborative framework in which ADown improves feature preservation during downsampling, CoordAtt strengthens spatial localization, and GCEM supplements global semantic context. These modules operate at different stages of the network and provide complementary enhancement from the feature-preservation, spatial-localization, and global-semantic perspectives. Rather than introducing redundant feature enhancement mechanisms, the three modules target different information dimensions and provide complementary improvements to the representation capability of ACG-YOLO while preserving its lightweight design.

3.2. Design Motivation and Collaborative Optimization Strategy

Although significant progress has been made in recent years in object detection technologies based on deep learning, the task of detecting violence and weapons continues to present specific challenges that differ from those of general-purpose object detection. Unlike conventional object detection, which focuses primarily on class classification, the detection of violence and weapons in public safety surveillance scenarios requires not only the accurate localization of dangerous weapons but also an understanding of the complex behavioral interactions between multiple individuals, whilst simultaneously meeting the stringent requirements for computational efficiency on edge devices in real-time surveillance systems. Consequently, this task essentially involves optimization requirements across three interrelated and equally critical dimensions: local detail perception, global semantic modeling and lightweight deployment.

The first challenge stems from the difficulty of detecting small-sized weapon targets. Dangerous items such as firearms and knives typically occupy only a tiny area of between tens and hundreds of pixels in surveillance footage, and are often obscured by people, affected by motion blur, and obscured by complex background textures. In traditional convolutional networks, during the downsampling process involving multiple-stride convolutions, high-frequency edge information and textural details attenuate layer by layer, resulting in the response signals for minute objects in deep feature maps becoming extremely weak or even disappearing entirely. Although increasing the input resolution or deepening the network can partially alleviate this problem, this would significantly increase the computational overhead, which directly conflicts with the need for lightweight edge deployment. Therefore, how to enhance the spatial localization capability for small-sized targets whilst ensuring the model remains compact has become the primary key issue in improving weapon detection performance.

The second challenge stems from the heavy reliance of violent behavior on modeling the overall context. The criteria for identifying violent behavior are not limited to local limb postures or single-target features, but depend on a holistic understanding of the complex spatial relationships and interaction patterns between multiple individuals. For example,

the same human posture may convey entirely different behavioral meanings on a crowded street and in an open square; relying solely on local receptive fields makes it difficult to accurately distinguish between behaviors. Although the Transformer architecture can effectively establish long-range dependencies through its self-attention mechanism, its computational complexity increases quadratically with the spatial dimensions of the feature map, making it difficult to meet the real-time inference requirements of edge devices when processing high-resolution surveillance feature maps. Consequently, there is an urgent need for a lightweight context-modeling strategy that balances global perception capabilities with computational efficiency, enabling the model to understand the semantic meaning of behaviors in complex interactive scenarios under limited resource constraints.

The third challenge stems from the strict constraints on computational resources in edge deployment environments. Public safety surveillance systems are typically deployed on resource-constrained platforms such as edge servers, embedded devices or low-power smart cameras, which impose stringent requirements on model parameter count, computational load and inference latency. In real-world monitoring scenarios, the limited computing power and power consumption constraints of edge devices mean that models must strike a realistic balance between accuracy and efficiency.

Although YOLOv11n provides a favorable accuracy–efficiency trade-off, its original architecture still presents several limitations for violence and weapon detection. Specifically, conventional downsampling may cause the loss of fine-grained information from small and partially occluded weapons; the lack of explicit coordinate-aware positional encoding may limit precise spatial localization; and the absence of a lightweight global semantic recalibration mechanism may restrict contextual feature enhancement in complex multi-person interactions. These limitations motivate the introduction of ADown, CoordAtt, and GCEM, respectively, to improve feature preservation, spatial localization, and global semantic representation, as illustrated in Figure 2.

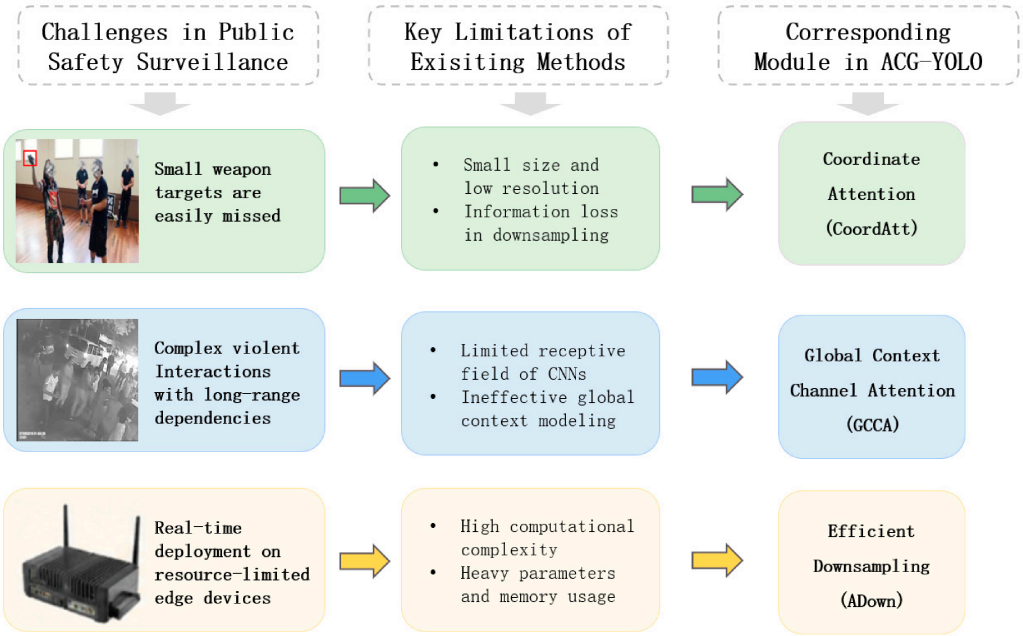


Figure 2. Motivation for the proposed ACG-YOLO framework.

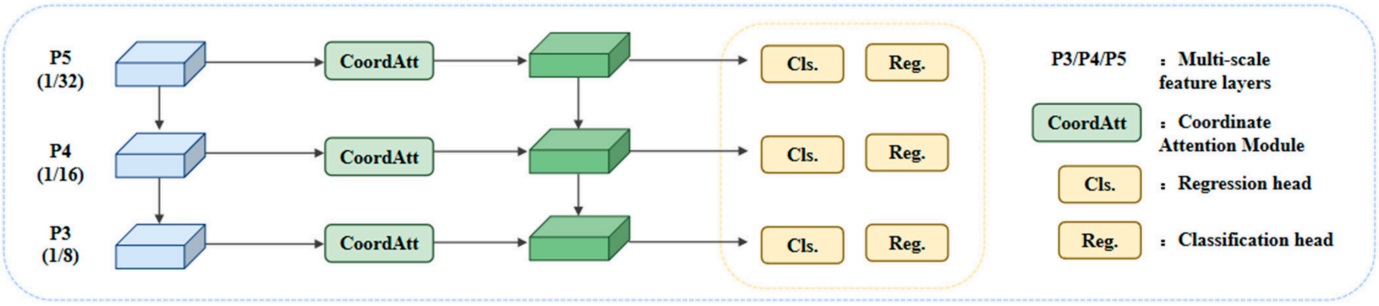
Based on the above analysis, this paper introduces ADown, Coordinate Attention (CoordAtt), and the lightweight Global Context Enhancement Module (GCEM) as targeted modifications to YOLOv11n. ADown improves the preservation of fine-grained information while reducing computational cost, CoordAtt enhances spatial localization through explicit

positional information, and GCEM strengthens global semantic representation through lightweight channel recalibration. The three modules therefore provide complementary improvements from the perspectives of feature preservation, spatial localization, and global semantic representation.

3.3. Spatial Localization Enhancement via Coordinate Attention

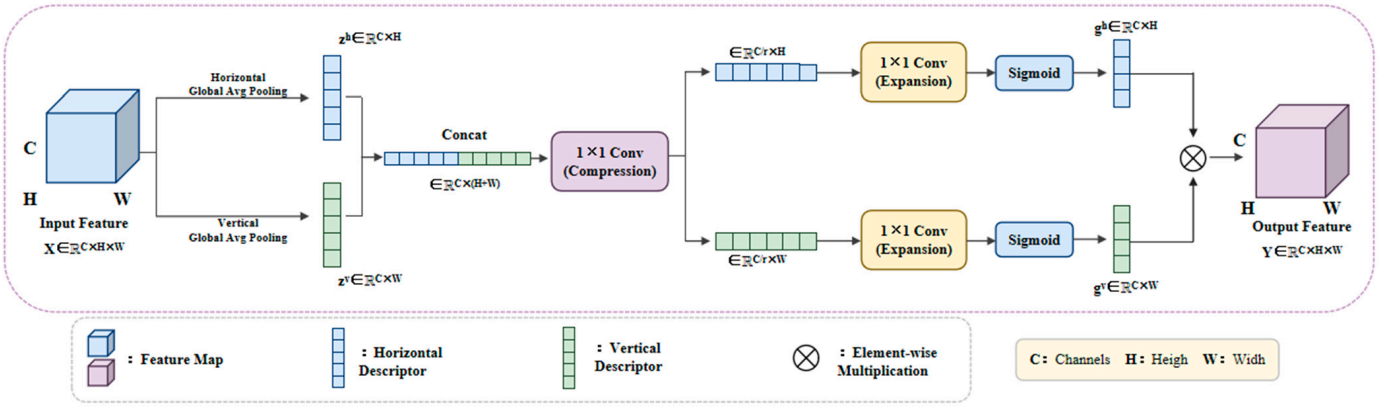
In public safety surveillance scenarios, dangerous weapon targets (such as knives and firearms) are typically characterised by their small size, limited textural information and susceptibility to partial occlusion by the human body. As convolutional networks continuously perform spatial downsampling through strided convolution and pooling operations, the high-frequency edge information and spatial position information retained in shallow layers gradually diminish with each layer. This results in extremely weak response signals for small targets in deep feature maps, making the model prone to missed detections or positioning errors. At the same time, the stringent requirements for model compactness in edge deployment scenarios mean that relying solely on increasing the input resolution or deepening the network to improve the detection of small targets is not a viable approach. Therefore, whilst maintaining the model's lightweight nature, enhancing the network's ability to localize targets can improve weapon detection performance while preserving the lightweight design.

Traditional channel attention mechanisms typically employ two-dimensional global average pooling to compress the entire spatial dimension into a single scalar, thereby modelling dependencies between channels. However, this 'global compression' strategy completely discards spatial position information during feature aggregation, preventing the attention weights from perceiving the specific location of the target, which is detrimental to precise localization in small-object detection tasks. In contrast, whilst the CBAM compensates to some extent for the lack of positional information through its spatial attention branch, its sequential spatial and channel attention operations introduce additional computational overhead and may overlap functionally with coordinate-based positional modeling. To address the aforementioned limitations, this paper introduces a Coordinate Attention mechanism into the multi-scale feature layers of the neck network, as shown in Figure 3. By explicitly embedding spatial coordinate information into the channel attention weights, it achieves a synergistic enhancement of both spatial localization capabilities and channel representation capabilities with minimal additional computational overhead.



(**a**)

Figure 3. *Cont.*



(**b**)

Figure 3. Deployment of Coordinate Attention in the multi-scale feature layers and its detailed structure. (**a**) Deployment of Coordinate Attention in the multi-scale feature layers of ACG-YOLO. (**b**) Structure of Coordinate Attention.

Specifically, given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, where C, H, and W denote the number of channels, height, and width, respectively, we first apply global average pooling along the horizontal and vertical directions, respectively, to produce two orientation-aware feature maps:

$$z_c^h(h) = \frac{1}{W}\sum_{i=0}^{W-1} x_c(h,i), \tag{1}$$

$$z_c^v(w) = \frac{1}{H}\sum_{j=0}^{H-1} x_c(j,w), \tag{2}$$

Here, $z_c^h \in \mathbb{R}^{C\times H}$ and $z_c^v \in \mathbb{R}^{C\times W}$ denote the horizontal and vertical orientation features, respectively, where h, j are index rows and i, w are index columns, with $x_c(h,i)$ representing the pixel value at the c-th channel, h-th row, and i-th column, and $x_c(j,w)$ defined analogously for the j-th row and w-th column. These two pooling operations aggregate global information along two orthogonal directions, enabling the subsequent attention computation to capture long-range dependencies along either the row or column dimension while preserving precise positional information. Subsequently, we concatenate z_c^h and z_c^v along the spatial dimension to form an intermediate feature map $f \in \mathbb{R}^{C\times(H+W)}$, followed by channel compression via a 1×1 convolution:

$$f = \delta\left(\mathrm{Conv}_{1\times1}\left(\left[z^h, z^v\right]\right)\right), \tag{3}$$

where $\left[z^h, z^v\right]$ denotes the feature concatenation operation along the spatial dimension, and δ represents the nonlinear activation function (specifically, SiLU in this work). Next, we split f along the spatial dimension into $f^h \in \mathbb{R}^{\frac{C}{r}\times H}$ and $f^v \in \mathbb{R}^{\frac{C}{r}\times W}$, where $\frac{C}{r}$ is the channel count of f and r is the reduction ratio. Each of these is then passed through a 1×1 convolution to restore the original channel number, followed by a Sigmoid activation to generate the attention weights:

$$g_c^h = \sigma\left(Conv1\times1\left(f^h\right)\right), g_c^v = \sigma(Conv1\times1(f^v)), \tag{4}$$

where σ denotes the Sigmoid function, and g^h and g^v are the horizontal and vertical attention weight maps, respectively. The final output feature map is then computed as:

$$Y = X \odot g^h \odot g^v, \tag{5}$$

where $\odot$ denotes element-wise multiplication, and g^h and g^v are broadcast along the horizontal and vertical dimensions, respectively, to match the spatial dimensions of X.

Through the aforementioned mechanism of bidirectional one-dimensional pooling and decoupled weight generation, coordinate attention enables the network to preserve precise positional information along both the row and column directions without incurring a significant increase in computational cost. This property proves particularly critical in detecting small-sized weapons such as knives and firearms: horizontal one-dimensional pooling captures the typically horizontal or slanted pose characteristics of weapons, while vertical pooling facilitates the discrimination of spatial positional relationships between the weapon and human limbs. Together, these operations help preserve edge and structural information, which can improve localization under complex background conditions.

It is worth noting that, within the collaborative optimization framework of ACG-YOLO, coordinate attention assumes the role of "spatial fine localization," whose core functionality can be succinctly characterized as answering "Where to Focus." Specifically, it enhances the network's spatial response to critical target regions by explicitly encoding spatial positional information. Complementing this role, the subsequently proposed GCEM in this paper centers on global semantic modeling with the objective of addressing "What to Focus," achieved via channel-wise global statistical recalibration to establish cross-regional behavioral semantic associations in complex violent scenarios. These two modules orthogonally enhance feature representation capability among the spatial and channel dimensions, respectively—coordinate attention operates on spatial structure, whereas GCEM acts upon channel semantics—each fulfilling its designated function without mutual interference. Together, they constitute the synergistic optimization mechanism of "spatial fine localization plus global semantic prior" within ACG-YOLO.

3.4. Lightweight Global Context Modeling Using GCEM

To address the lack of effective global semantic modeling in complex multi-person interaction scenarios for violence detection, this paper proposes a lightweight global context enhancement strategy, termed the Global Context Enhancement Module (GCEM). Unlike conventional channel attention mechanisms, which are primarily designed for feature importance screening in category discrimination tasks, GCEM aims to leverage low-cost global statistical information to provide a semantic prior for behavior understanding in surveillance scenarios. Its core philosophy is not to construct a novel attention computation paradigm, but rather to introduce global scene information into the convolutional detection network under lightweight constraints, thereby enhancing the model's capability to comprehend cross-regional behavioral relationships and complex interaction patterns. The architecture of GCEM is illustrated in Figure 4. The module is designed as a task-oriented lightweight adaptation of the squeeze-and-excitation (SE) principle, with emphasis on global contextual representation under computational constraints. GCEM first extracts global channel statistics through global average pooling, followed by bottleneck-based cross-channel transformation and channel recalibration, and semantic enhancement with an approximate global receptive field is ultimately achieved via channel recalibration. The channel attention generation operates on a compact global descriptor rather than the full spatial feature map, thereby avoiding explicit spatial similarity computation and pairwise spatial interaction. This enables GCEM to provide enhanced feature maps with a global semantic prior to spatial localization modules such as CoordAtt, while maintaining a relatively low computational overhead.

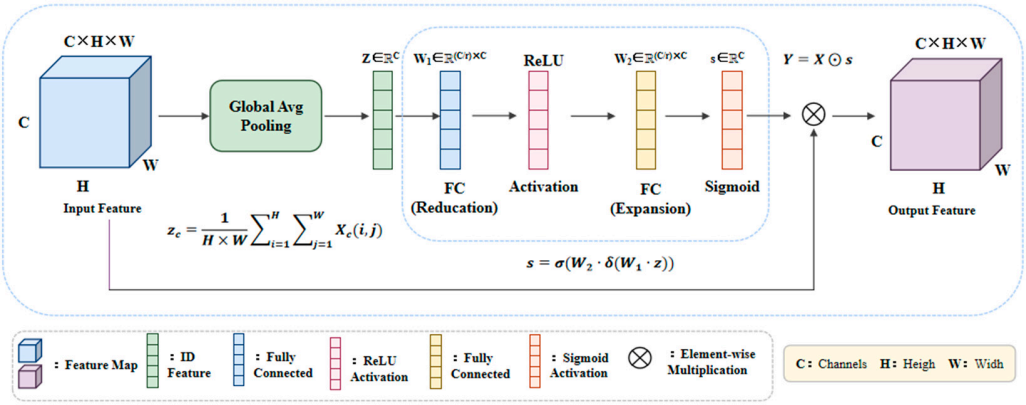


Figure 4. Structure of the proposed GCEM.

Let the input feature map be denoted as $X \in \mathbb{R}^{C \times H \times W}$, where C, H, and W denote the number of channels, height, and width, respectively. We first perform global context extraction by compressing the spatial dimensions via global average pooling (GAP) to extract the global statistical information for each channel:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X_c(i,j), z \in \mathbb{R}^C, \tag{6}$$

where $X_c(i,j)$ denotes the feature value at position (i,j) in the c-th channel. This operation aggregates spatial information into a channel-wise global descriptor, enabling subsequent attention computation to be performed based on the global context. Subsequently, channel attention recalibration is performed, where the global descriptor z is fed into a lightweight bottleneck structure to generate channel-wise attention weights. This bottleneck structure comprises two fully connected layers, which enable cross-channel information interaction via dimensionality reduction and subsequent expansion:

$$s = \sigma(W_2 \cdot \delta(W_1 \cdot z)), \tag{7}$$

where $W_1 \in \mathbb{R}^{(\frac{C}{r}) \times C}$ denotes the dimensionality-reduction linear transformation (with reduction ratio $r = 16$), and $W_2 \in \mathbb{R}^{C \times (\frac{C}{r})}$ the corresponding expansion linear transformation, with $r = 16$ being the reduction ratio, δ the ReLU activation function, and σ the Sigmoid activation function. The dimensionality-reduction operation compresses the channel count to $\frac{C}{r}$, effectively reducing the parameter overhead, while the expansion operation restores it to the original number of channels, yielding the final channel attention weights $s \in \mathbb{R}^C$. Finally, the resulting channel attention weights s are multiplied element-wise with the original feature map X along the channel dimension to perform feature recalibration:

$$Y = X \odot s, \tag{8}$$

where $\odot$ denotes element-wise multiplication, and s is broadcast along the spatial dimensions to match the size of X. This operation enables the network to adaptively enhance the feature responses of critical channels while suppressing interference from irrelevant ones.

In terms of computational complexity, the global average pooling operation requires $O(CHW)$ computations to aggregate the spatial information of all C channels. The two fully connected transformations in the bottleneck structure have dimensions $\frac{C}{r} \times C$ and $C \times \frac{C}{r}$, respectively, resulting in a computational cost of $O(\frac{C^2}{r})$ in total. The subsequent

channel-wise feature recalibration requires another $O(CHW)$ element-wise operations. Therefore, the overall computational complexity of GCEM can be characterized as:

$$O(CHW + \frac{C^2}{r}), \tag{9}$$

The computational cost of GCEM consists of global average pooling with complexity proportional to CHW and bottleneck transformations with complexity proportional to $\frac{C^2}{r}$. Thus, the overall complexity can be expressed as $O\left(CHW + \frac{C^2}{r}\right)$. Moreover, GCEM does not introduce spatial similarity matrices or pairwise spatial interactions, thereby maintaining a relatively low computational overhead while providing global contextual information for subsequent spatial localization modules such as CoordAtt.

It should be clearly noted that GCEM structurally inherits the classic SE architecture; its core contribution lies not in proposing a new attention computation paradigm, but in task adaptation and system-level collaborative optimization. Although GCEM shares the global pooling operation with SE, their motivations and applications are fundamentally distinct. SE is designed for generic feature recalibration, whereas GCEM is specifically tailored for surveillance violence understanding, emphasizing the decisive role of global scene context in distinguishing normal interactions from violent acts. Compared with spatial self-attention, GCEM has lower computational requirements because it operates on a compact global descriptor rather than on pairwise spatial interactions. In terms of computational efficiency, GCEM avoids the explicit computation of spatial similarity matrices and pairwise spatial interactions. Its channel attention generation operates on a compact global descriptor, while the bottleneck transformation reduces the cost of cross-channel interaction. As a result, GCEM provides global contextual modeling with relatively low computational overhead compared with spatial self-attention mechanisms. In terms of feature representation, violence recognition relies more on "which channels carry behavior-relevant scene information" than on "spatial location interactions," and GCEM's channel recalibration mechanism precisely aligns with this requirement. Furthermore, GCEM and Coordinate Attention (CA) exhibit a clear orthogonal complementary relationship: CA concentrates on "where the target is," enhancing spatial localization via horizontal and vertical one-dimensional pooling, which is particularly beneficial for small-sized weapon detection, whereas GCEM focuses on "what semantic meaning the target conveys," identifying behavior-discriminative critical channels through channel-wise statistics. The two modules work synergistically within ACG-YOLO, respectively reinforcing spatial structure and channel semantics without mutual interference, thereby forming a dual-dimensional synergistic enhancement mechanism of "spatial fine localization plus global semantic prior," which collectively improves the robustness of violence and weapon detection.

3.5. Efficient Feature Compression via ADown

Downsampling is a core operation in convolutional neural networks for enlarging the receptive field and reducing spatial resolution. However, both strided convolution and max pooling may cause information loss: strided convolution reduces spatial resolution through subsampling, whereas max pooling retains only the maximum response within each pooling window. In violence and weapon detection, small targets such as knives and firearms rely heavily on contours and local textures. Once these fine-grained features are weakened, the resulting deep-layer responses become extremely faint, making it difficult for subsequent modules to recover detection capability. Meanwhile, aggressive compression tends to cause small targets to vanish, whereas conservative strategies lead to uncontrolled computational costs, making the trade-off between fidelity and efficiency particularly acute in edge deployment scenarios. To address this, we replace all standard

strided-convolution downsampling operations in the backbone network with the ADown module, the architecture of which is illustrated in Figure 5. Unlike conventional single-path downsampling, ADown adopts a parallel dual-branch feature compression strategy, performing the feature downsampling process via the collaborative operation of strided convolution and max pooling. This design helps retain complementary high-frequency and locally salient information while reducing spatial resolution.

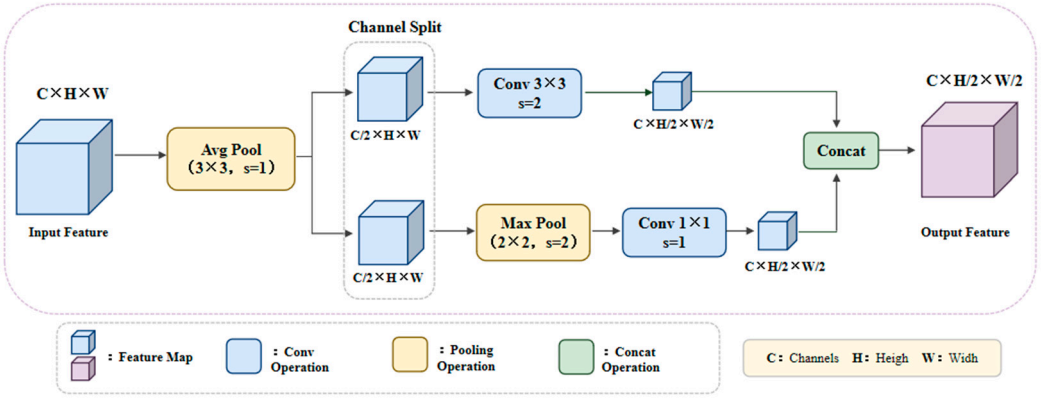


Figure 5. Structure of the ADown module.

Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, where C, H, and W denote the number of channels, height, and width, respectively. ADown operates via two parallel branches. The left branch is a strided convolution branch, which applies a standard convolutional downsampling operation to the input feature map using a 3×3 kernel with a stride of 2. This branch is capable of extracting high-frequency detail information, albeit at the cost of potentially losing some salient features.

$$X_{\text{conv}} = \text{Conv}_{3\times 3, s=2}(X), \tag{10}$$

The right branch is a pooling–convolution branch, which first applies 2×2 max pooling to preserve locally salient features, followed by a 1×1 convolution to adjust the channel dimension:

$$X_{\text{pool}} = \text{Conv}_{1\times 1}(\text{MaxPool}_{2\times 2}(X)), \tag{11}$$

The output feature maps from the two branches are concatenated along the channel dimension, yielding a feature map with $2C$ channels:

$$X_{\text{cat}} = [X_{\text{conv}}, X_{\text{pool}}] \in \mathbb{R}^{2C\times \frac{H}{2}\times \frac{W}{2}}, \tag{12}$$

Finally, a 1×1 convolution is applied to compress the channel count back to C, achieving feature fusion:

$$X_{\text{out}} = \text{Conv}_{1\times 1}(X_{\text{cat}}) \in \mathbb{R}^{C\times \frac{H}{2}\times \frac{W}{2}}, \tag{13}$$

This concatenation operation endows the final output features with both the high-frequency detail extraction capability from the convolution branch and the locally salient information preservation capability from the pooling branch, rendering the two branches complementary rather than redundant. It is worth emphasizing that the ADown module introduces only a minimal increase in parameter count—both branches employ small kernel sizes (3×3 and 1×1) with the channel numbers halved in each—thereby effectively controlling the overall model's parameter and computational overhead while maintaining effective feature representation.

From a task adaptation perspective, ADown assumes the fundamental role of "efficiency assurance" within ACG-YOLO, with its core contribution lying in providing subsequent modules with feature inputs that incur less information loss, rather than directly enhancing representational capacity. Coordinate Attention relies on precise positional information for spatial localization, while GCEM depends on rich channel-wise features for global semantic modeling—both of which presuppose the effective preservation of information during the downsampling stage. Should downsampling compromise edge contours or attenuate channel responses, the subsequent modules would struggle to recover the lost information. The three modules are clearly delineated in their respective roles: ADown safeguards information integrity during feature compression, CoordAtt reinforces spatial localization, and GCEM supplies global semantic understanding. Together, these modules address complementary aspects of feature preservation, spatial localization, and global context modeling, forming a lightweight framework for violence and weapon detection.

4. Results

4.1. Implementation Details and Datasets

All computational experiments in this study were conducted on the Ubuntu 20.04 LTS operating system (Canonical Ltd., London, UK), with Python 3.8 (Python Software Foundation, Wilmington, DE, USA) as the programming language, and CUDA 11.8 (NVIDIA Corporation, Santa Clara, CA, USA) along with PyTorch 2.1.2 (Meta Platforms, Inc., Menlo Park, CA, USA) as the deep learning framework. The hardware configuration comprised an NVIDIA vGPU with 32 GB of memory (NVIDIA Corporation, Santa Clara, CA, USA) and an Intel Xeon Platinum 8470Q CPU (16 vCPU cores; Intel Corporation, Santa Clara, CA, USA). Model training followed the standard configuration strategy of Ultralytics YOLO11 (Version 8.3.0, Ultralytics Inc., Los Angeles, CA, USA): input images were uniformly resized to 640 $\times$ 640 pixels; training was conducted for a total of 300 epochs with a batch size of 16. The optimizer was automatically selected by the framework, with the initial learning rate set to 0.01 and annealed to 0.0001 following a cosine annealing schedule. Data augmentation included Mosaic augmentation, MixUp, random affine transformations (rotation, translation, scaling, and shearing), HSV color-space augmentation, horizontal flipping, and random erasing. These augmentation strategies were applied to improve model robustness under diverse surveillance conditions such as varying illumination, occlusion, and viewpoint changes. All training experiments were performed on the same vGPU platform to ensure reproducibility and fair comparison. The edge-deployment potential was evaluated through model complexity metrics (parameters and GFLOPs) and inference FPS, which serve as standard proxies for assessing edge suitability in the literature.

To comprehensively evaluate the performance of the proposed model, this study employs two distinct public datasets: COCO2017 and a dedicated Violence and Weapon Detection dataset. The Violence and Weapon Detection dataset is designed for violence and weapon detection, comprising 13,636 images categorized into four classes: NonViolence, Violence, Guns, and Knife. The dataset is split into training, validation, and test sets with 12,266, 948, and 422 images, respectively. The total number of annotated object instances is 16,571, among which 4647 belong to NonViolence, 5820 to Violence, 2859 to Guns, and 3245 to Knife. Weapon instances (Guns and Knife) account for 6104 instances, representing 36.8% of all annotations. All images are uniformly resized to 640 $\times$ 640 pixels during training, with the original images maintaining the same resolution. The dataset encompasses real-world scenarios across varying times of day, diverse illumination conditions, and different cultural contexts, making it suitable for evaluating models for violence and weapon detection under diverse surveillance conditions. We additionally adopt the COCO2017 dataset to validate

the model's generalization capability. This dataset comprises 80 object categories, with 118,287 images in the training set and 5000 images in the validation set.

4.2. Evaluation Metrics

We evaluate ACG-YOLO using standard object detection metrics, including Precision, Recall, mean Average Precision (mAP), F1-score, parameter count (Params), computational complexity (GFLOPs), and frames per second (FPS). Precision and Recall are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{14}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{15}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. AP measures the detection performance of an individual class and is calculated from its Precision–Recall curve:

$$\text{AP} = \int_0^1 \text{P(R)dR}, \tag{16}$$

Mean Average Precision (mAP), in turn, denotes the average of AP values across all categories, and is computed as follows:

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N} \text{AP}_i, \tag{17}$$

where N is the total number of categories. In this work, mAP@0.5 and mAP@0.5:0.95 denote the mean AP at an IoU threshold of 0.5 and over IoU thresholds from 0.5 to 0.95 with a step of 0.05, respectively. The F1-score is the harmonic mean of Precision and Recall, serving as a comprehensive metric for evaluating the detection performance of the model, and is computed as follows:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{18}$$

Params and GFLOPs characterize model size and computational cost, while FPS reflects inference speed and real-time capability. Together, these metrics characterize detection performance, model complexity, and inference efficiency.

4.3. Comparison with Representative and Mainstream Lightweight Detectors

To comprehensively evaluate the performance of the proposed method, we systematically compare ACG-YOLO with several current mainstream lightweight object detection models, including YOLOv5n [38], YOLOv8n [39], YOLOv10n [40], and YOLOv11n [41], as well as representative methods of different architectures such as SSD, RT-DETR [42], PP-YOLOE [43], and YOLO-NAS [44]. All models are trained and evaluated under identical data splits, training strategies, and evaluation metrics to ensure the fairness and reliability of the comparative results. Table 1 presents the experimental results of different models on the Violence and Weapon Detection dataset.

To further evaluate the model's performance on weapon-specific categories, we report class-wise AP for Guns and Knife, as shown in Table 2.

Table 1. Performance comparison of representative object detection models on the Violence and Weapon Detection dataset.

Methods	map@0.5	map@0.5:0.95	Precision	Recall	Parameters/M	GFLOPs	FPS	F1-Score
YOLOv5n	0.861	0.594	0.84	0.84	2.18	5.8	749.5	0.839
YOLOv8n	0.884	0.633	0.897	0.835	2.69	6.8	764.92	0.864
YOLOv10n	0.887	0.641	0.889	0.822	2.70	8.2	608.52	0.854
YOLOv11n	0.863	0.634	0.858	0.823	2.58	6.3	637.93	0.838
SSD	0.674	0.415	-	-	24.15	30.58	249.97	-
RT-DETR	0.724	0.475	0.822	0.679	19.88	57	80.29	0.743
PP-YOLOE	0.856	-	-	-	7.65	17.36	66.08	-
YOLO-NAS	0.721	0.515	0.839	0.840	19.02	-	102.84	0.839
Ours	0.892	0.639	0.885	0.842	2.07	5.1	504.62	0.861

Table 2. Class-wise AP comparison for weapon categories between YOLOv11n and ACG-YOLO.

Method	Class	map@0.5	map@0.5:0.95
YOLOv11n	Guns	0.968	0.796
YOLOv11n	Knife	0.666	0.383
ACG-YOLO	Guns	0.968	0.783
ACG-YOLO	Knife	0.771	0.401

For Guns, ACG-YOLO maintains virtually identical detection performance to the baseline, demonstrating that the proposed lightweight modifications do not compromise the already strong detection capability for this category. More importantly, for Knife—which exhibits lower baseline performance due to its slender shape and susceptibility to occlusion—ACG-YOLO achieves a substantial improvement of 10.5 percentage points in mAP50, confirming that the collaborative optimization of spatial localization (CoordAtt) and global semantic modeling (GCEM) effectively enhances detection accuracy for challenging weapon categories in complex surveillance scenarios.

From an overall performance perspective, ACG-YOLO achieves a favorable comprehensive balance among detection accuracy, model complexity, and inference efficiency. Compared to the baseline YOLOv11n, ACG-YOLO improves the mAP@0.5 from 86.3% to 89.2%, yielding a gain of 2.9 percentage points; reduces the parameter count from 2.58 M to 2.07 M, corresponding to a 19.9% reduction; and decreases the computational cost from 6.3 GFLOPs to 5.1 GFLOPs, representing a 19.0% decrease. These results demonstrate a simultaneous optimization of accuracy and efficiency.

In terms of detection accuracy, ACG-YOLO achieves an mAP@0.5 of 89.2%, outperforming YOLOv8n, YOLOv10n, and the baseline YOLOv11n. It also attains an mAP@0.5:0.95 of 63.9%, surpassing all compared models, which indicates that the model maintains stable performance under stringent localization requirements—a property that is particularly critical for the precise localization of hazardous weapons in public safety surveillance. The Precision and Recall are 88.5% and 84.2%, respectively, achieving the highest Recall while maintaining a relatively high Precision. This demonstrates that the model strikes a better balance between reducing false positives and suppressing false negatives. By contrast, although YOLOv5n has a lower parameter count, its mAP@0.5:0.95 is substantially lower than that of ACG-YOLO, suggesting that merely compressing parameters may lead to degraded localization accuracy. In this regard, the spatial position encoding of CoordAtt effectively compensates for this shortcoming.

In terms of model efficiency, ACG-YOLO achieves a parameter count of only 2.07 M and a computational cost of merely 5.1 GFLOPs, both of which are the lowest among all compared models. Compared to YOLOv8n, the parameter count and computational

cost are reduced by 23.0% and 25.0%, respectively; against YOLOv10n, the corresponding reductions are 23.3% and 37.8%; and relative to PP-YOLOE and YOLO-NAS, both metrics are reduced by over 70%. This efficiency advantage is primarily attributable to the dual-branch parallel downsampling strategy of the ADown module. In comparison with RT-DETR, ACG-YOLO achieves a 16.8 percentage-point higher mAP@0.5, while reducing the parameter count and computational cost by 89.6% and 91.2%, respectively. These results highlight the importance of lightweight design for resource-constrained deployment.

From a deployment perspective, public safety surveillance systems are typically deployed on resource-constrained platforms, where models must balance detection accuracy with real-time inference requirements. Considering multi-dimensional metrics, ACG-YOLO provides a favorable accuracy–efficiency trade-off, achieving the highest mAP@0.5 with the lowest GFLOPs among the compared models. Compared to YOLOv8n and YOLOv10n (which offer comparable accuracy but incur higher computational overhead) as well as YOLOv5n (which has lower computational cost but insufficient accuracy), ACG-YOLO strikes a more desirable balance. The experimental results thoroughly validate the effectiveness of the progressive synergistic optimization strategy, namely "efficiency assurance → spatial fine localization → global semantic prior." They further demonstrate that ACG-YOLO, while maintaining a lightweight design, effectively alleviates the issue of insufficient feature representation capability commonly encountered in lightweight networks, rendering it more suitable for edge deployment in real-time public safety surveillance scenarios.

4.4. Modular and Framework-Level Evaluation of GCEM

In violence and weapon detection tasks, the choice of attention mechanism affects both feature representation and computational efficiency. To comprehensively evaluate the effectiveness of GCEM in violence and weapon detection and its adaptability within the ACG-YOLO framework, we design two progressive comparative experiments. The first experiment (Section 4.4.1) compares the standalone performance of multiple typical attention mechanisms under the same baseline network, to evaluate the standalone effectiveness of GCEM. The second experiment (Section 4.4.2) integrates each attention mechanism into the complete framework separately, addressing the question of "whether GCEM is the most suitable for ACG-YOLO." Together, these two experiments form a progressive validation chain of "module-level verification → framework-level verification," providing a comprehensive assessment of the practical value of GCEM.

4.4.1. Performance Comparison of Different Attention Mechanisms

To validate the effectiveness of GCEM in violence and weapon detection, we first compare the performance of several typical lightweight attention mechanisms under the same baseline network (YOLOv11n) and identical training configurations, including SE, CBAM, ECA, GAM (Global Attention Mechanism) [45], EMA (Efficient Multi-Scale Attention) [46], and the proposed GCEM. All methods are tested under the same training strategies and experimental configurations to ensure the fairness of the experimental results. Table 3 reports the experimental results of different attention mechanisms on the Violence and Weapon Detection dataset.

Table 3. Performance comparison of different attention mechanisms on the Violence and Weapon Detection dataset.

Method	map@0.5	map@0.5:0.95	Params (M)	GFLOPs
YOLOv11n	0.863	0.634	2.58	6.3
+SE	0.879	0.636	2.59	6.3
+CBMA	0.889	0.645	2.65	6.3
+GAM	0.877	0.640	4.22	7.6
+ECA	0.886	0.644	2.58	6.3
+EMA	0.881	0.636	2.59	6.4
+GCEM (Ours)	0.883	0.637	2.42	6.1

As shown in Table 3, the incorporation of different attention mechanisms consistently improves the detection performance over the baseline YOLOv11n to varying degrees. However, these methods exhibit distinct trade-offs among detection accuracy, model complexity, and computational overhead. Compared with the baseline model, all evaluated attention mechanisms improve at least one of the reported detection metrics, indicating that attention mechanisms play a positive role in violence and weapon detection tasks.

Specifically, SE enhances feature representation through channel recalibration; however, its improvement in localizing small targets is limited due to the loss of spatial positional information during global pooling. CBAM, which combines channel and spatial attention, achieves the highest detection accuracy among all compared methods, yet its additional spatial attention branch incurs increased model parameters and computational overhead. GAM further strengthens cross-dimensional feature interaction, but the substantial increase in both parameter count and computational cost renders it less favorable for edge deployment. EMA attains stable performance while maintaining low complexity; nevertheless, its multi-branch structure may introduce additional memory access overhead during deployment. ECA achieves competitive detection accuracy via lightweight feature enhancement using local cross-channel convolution with almost no increase in complexity; however, its local channel interaction mechanism still struggles to fully model the global semantic associations inherent in complex violent scenes.

In contrast, GCEM achieves 88.3% mAP@0.5 and 63.7% mAP@0.5:0.95 while maintaining the lowest model complexity of 2.42 M parameters and 6.1 GFLOPs, improving over the baseline by 2.0 and 0.3 percentage points, respectively. Although its detection accuracy when deployed alone is slightly lower than that of CBAM and ECA, GCEM reduces the parameter count and computational cost by approximately 8.7% and 3.2%, respectively, relative to CBAM, while incurring only a 0.6 percentage-point loss in mAP@0.5. Relative to ECA, it achieves further reductions in model complexity with a negligible accuracy gap of only 0.3 percentage points, indicating favorable parameter efficiency. It is worth noting that GCEM adopts a lightweight bottleneck architecture and is integrated via holistic network-level substitution, which not only avoids increasing model complexity but actually reduces both parameter count and computational overhead, making it more suitable for resource-constrained edge deployment scenarios.

The above results indicate that global contextual information is of significant importance for violence and hazardous weapon detection. Unlike generic object detection tasks, violence recognition requires simultaneous understanding of multi-person interactions and scene semantics, while hazardous weapon targets are characterized by small sizes and susceptibility to occlusion. Relying solely on local convolutional receptive fields or local channel interactions is insufficient for adequately modeling such complex relationships. GCEM captures scene-level semantic statistics via global average pooling and leverages lightweight channel recalibration to establish global semantic associations, thereby en-

hancing the network's semantic representation capacity while maintaining a relatively low computational overhead. The experimental results further support this efficiency advantage, with GCEM achieving the lowest parameter count and GFLOPs among the compared attention mechanisms. It should be noted, however, that a single attention module achieving high accuracy when deployed independently does not necessarily guarantee its optimal performance within the complete detection framework. Different attention mechanisms may exhibit varying degrees of complementarity or redundancy with modules such as CoordAtt and ADown; consequently, it is difficult to comprehensively evaluate their practical value based solely on single-module experiments. The following section will further compare the integration effects of different attention mechanisms within the complete ACG-YOLO framework, aiming to analyze the synergistic optimization capabilities among the modules.

4.4.2. Performance Comparison Under the Complete Framework

The experiments in the preceding subsection have validated the effectiveness of GCEM as a standalone attention module. In an actual detection framework, however, the ultimate value of a module depends not only on its standalone performance but also on its collaborative compatibility and system-level integration effectiveness with other modules. As previously discussed, the detection accuracy of ACG-YOLO does not stem from the contribution of a single module, but rather from the synergistic enhancement among ADown, CoordAtt, and the global context modeling module. Therefore, the accuracy advantage of a single attention mechanism when deployed independently does not necessarily imply that it will achieve optimal performance within the complete framework—different attention mechanisms, with their distinct spatial/channel modeling strategies, may introduce varying degrees of redundancy or conflict with CoordAtt, thereby affecting the final effectiveness of system integration. To comprehensively evaluate the actual contribution of each attention mechanism within the complete ACG-YOLO framework, we integrate CBAM, ECA, and GCEM with ADown and CoordAtt, respectively, forming three combined schemes for comparative experiments. The results are presented in Table 4.

Table 4. Performance comparison of different attention mechanisms under the complete ACG-YOLO framework.

Method	map@0.5	map@0.5:0.95	Params (M)	GFLOPs
ADown + CoordAtt + CBAM	0.885	0.647	2.26	5.3
ADown + CoordAtt + ECA	0.886	0.640	2.07	5.1
ADown + CoordAtt + GCEM	0.892	0.639	2.07	5.1

Table 4 further compares the integration effects of different attention mechanisms within the complete ACG-YOLO framework. Compared with the standalone deployment experiments in Table 3, the results show that standalone performance does not necessarily predict performance within the complete framework. In the standalone experiments, CBAM and ECA achieve mAP@0.5 of 88.9% and 88.6%, respectively, both higher than GCEM's 88.3%. In the complete framework, however, the GCEM variant attains the highest mAP@0.5 of 89.2%, surpassing CBAM's 88.5% and ECA's 88.6%. This result suggests that module compatibility can be an important factor in determining overall framework performance, beyond the standalone effectiveness of an individual attention module.

The primary reason for this phenomenon lies in the differing synergistic patterns between various attention mechanisms and CoordAtt. CBAM simultaneously introduces both spatial and channel attention, with its spatial attention branch exhibiting a certain degree of functional overlap with CoordAtt's positional encoding, which tends to result in

redundant modeling of spatial features. ECA, on the other hand, focuses on local channel interactions and struggles to establish effective global semantic complementarity with CoordAtt. In contrast, GCEM generates channel-wise semantic enhancement from global statistical information, while CoordAtt is responsible for spatial positional encoding. The two modules correspond to "Where to Focus" and "What to Focus," respectively, providing complementary spatial and channel-wise feature enhancement within the framework.

In terms of model complexity, the GCEM variant maintains the same parameter count (2.07 M) and computational cost (5.1 GFLOPs) as the ECA variant, both of which are lower than those of the CBAM variant (2.26 M and 5.3 GFLOPs). This indicates that GCEM achieves improved detection performance without incurring additional computational overhead, thereby preserving the model's lightweight advantage.

By synthesizing the results in Tables 2 and 3, it can be observed that the advantage of GCEM does not lie in achieving the highest detection accuracy as a standalone module, but rather in exhibiting superior collaborative capability within the complete framework. By establishing complementary functional divisions with CoordAtt and ADown, GCEM achieves a favorable accuracy–efficiency trade-off while maintaining a lightweight design, further validating its rationale as the global context modeling module for ACG-YOLO.

4.5. Ablation Study

To systematically validate the effectiveness and synergistic effects of each module, we conduct a comprehensive ablation study using YOLOv11n as the baseline. The three modules—ADown, CoordAtt, and GCEM—are progressively integrated into eight configurations through controlled experimental designs, as detailed in Table 5. The results show that each module provides an individual performance contribution, the pairwise combinations further improve detection performance, and the full model provides the best overall balance between accuracy and computational efficiency.

Table 5. Ablation test results. "√" indicates that the corresponding module is included in the model, and "×" indicates that it is not included.

Method	ADown	CoordAtt	GCEM	map@0.5	map@0.5:0.95	Precision/%	Recall	Parameters/M	GFLOPs	FPS
Baseline	×	×	×	0.863	0.634	0.868	0.823	2.58	6.3	637.93
A	√	×	×	0.875	0.640	0.885	0.836	2.10	5.1	561.6
B	×	√	×	0.879	0.644	0.868	0.836	2.59	6.3	552.32
C	×	×	√	0.883	0.637	0.902	0.827	2.42	6.1	666.71
A+B	√	√	×	0.881	0.645	0.874	0.833	2.07	5.1	538.98
A+C	√	×	√	0.890	0.650	0.900	0.836	2.07	5.1	592.13
B+C	×	√	√	0.891	0.650	0.886	0.849	2.42	6.1	510.47
Ours-full	√	√	√	0.892	0.639	0.885	0.842	2.07	5.1	504.62

Table 5 presents the ablation experimental results for each module. From the single-module experiments (A, B, C), it can be observed that ADown, CoordAtt, and GCEM each independently improve model performance, albeit with different emphases. GCEM brings the largest improvement in detection accuracy (mAP@0.5 rising from 86.3% to 88.3%), indicating that global context modeling contributes to violence and weapon detection performance. CoordAtt raises mAP@0.5 to 87.9%, confirming that explicit positional encoding effectively enhances small-target localization capability. ADown, although yielding a relatively modest accuracy gain (87.5%), reduces parameter count and computational cost by 18.6% and 19.0%, respectively, suggesting that its main contribution is to reduce model complexity while providing feature representations for subsequent processing.

A further comparison of the two-module combinations (A+B, A+C, and B+C) reveals that the different modules exhibit clear complementary effects. Among these, A+C and

B+C achieve mAP@0.5 of 89.0% and 89.1%, respectively, both substantially outperforming their corresponding single-module results, indicating that the combination of GCEM with ADown and CoordAtt further unleashes the capability of global semantic modeling. In contrast, the combination of ADown and CoordAtt alone yields a detection accuracy of 88.1%, with a relatively limited improvement, suggesting that the enhancement of spatial localization capability still requires the assistance of global semantic information to adequately adapt to the behavioral discrimination demands in complex violent scenarios.

When all three modules are integrated simultaneously, the full model achieves the best overall performance, with mAP@0.5 reaching 89.2%—a 2.9 percentage-point improvement over the baseline—while the parameter count decreases from 2.58 M to 2.07 M and GFLOPs from 6.3 to 5.1, corresponding to reductions of 19.8% and 19.0%, respectively. Although the full model achieves only a marginal accuracy gain of 0.1 percentage points over the best two-module combination (B+C), it further improves overall performance while maintaining the lightweight advantage, indicating that the three modules can form an effective synergistic optimization mechanism. Specifically, ADown provides an efficient feature representation with richer information retention, CoordAtt reinforces spatial localization capability, and GCEM undertakes global semantic modeling. Together, they complement each other across three dimensions—computational efficiency, spatial information, and semantic representation—collectively achieving the optimal accuracy–efficiency trade-off.

The above ablation experiments demonstrate that the performance improvement of ACG-YOLO does not originate from any single module, but rather from the synergistic optimization of ADown, CoordAtt, and GCEM across three dimensions: computational efficiency, spatial localization, and global semantics. This further validates the effectiveness of the proposed collaborative design framework.

4.6. Hyperparameter Sensitivity Analysis

To evaluate the robustness of ACG-YOLO with respect to the principal hyperparameters of the proposed modules, we conduct a sensitivity analysis on the reduction ratio r used in both CoordAtt and GCEM. The reduction ratio controls the degree of channel compression in the bottleneck structure and directly influences the trade-off between model complexity and representation capability. We train three model variants with r = 8, 16, and 32 under identical experimental conditions (200 epochs, same data split, same optimizer settings), and report the results in Table 6.

Table 6. Sensitivity analysis of the reduction ratio r in CoordAtt and GCEM.

r	map@0.5	map@0.5:0.95	Params (M)	GFLOPs
8	0.848	0.607	2.11	5.2
16	0.868	0.622	2.07	5.1
32	0.876	0.625	2.08	5.2

As shown in Table 6, ACG-YOLO maintains stable detection performance across the tested range of reduction ratios. The variation in mAP@0.5:0.95 is only 1.8 percentage points (from 0.607 to 0.625), indicating that the model's performance is not highly sensitive to this hyperparameter. In terms of model complexity, the parameter count varies by less than 2% across the three configurations, while the computational cost remains nearly constant (5.1–5.2 GFLOPs). Notably, the default configuration r = 16 achieves the lowest parameter count (2.07 M) and computational cost (5.1 GFLOPs) while maintaining detection accuracy comparable to r = 32 (0.868 vs. 0.876 mAP@0.5), demonstrating a favorable accuracy–efficiency trade-off. These results confirm that the proposed framework exhibits favorable

robustness to the reduction ratio, and the default setting of r = 16 represents a reasonable and stable choice that balances model complexity and detection accuracy.

4.7. Generalization Evaluation

To further validate the generalization capability of the proposed method, we retrain and test both the baseline YOLOv11n and the improved ACG-YOLO on the COCO2017 dataset. The experimental results are presented in Table 7.

Table 7. Generalization results on the COCO2017 dataset.

Methods	map@0.5	map@0.5:0.95	Precision	Recall	Parameters/M	GFLOPs	F1-Score
YOLOv11n	0.863	0.661	0.831	0.762	2.62	6.5	0.7953
ACG-YOLO	0.896	0.676	0.869	0.794	2.10	5.2	0.8299

As shown in Table 7, ACG-YOLO also improves the reported detection metrics on the COCO2017 dataset. Compared to the baseline YOLOv11n, the improved model raises mAP@0.5 from 86.3% to 89.6% and mAP@0.5:0.95 from 66.1% to 67.6%, with consistent gains observed in Precision, Recall, and F1-score. Meanwhile, the parameter count decreases from 2.62 M to 2.10 M and GFLOPs from 6.5 to 5.2, indicating that the proposed method not only improves detection accuracy but also further reduces model complexity, achieving a favorable accuracy–efficiency balance.

It is worth noting that, despite the notable differences between COCO2017 and the Violence and Weapon Detection dataset in terms of object categories, scene distributions, and target scales, ACG-YOLO consistently achieves performance improvements on both datasets. These results suggest that the performance gains are not limited to the Violence and Weapon Detection dataset, but rather effectively enhances the model's feature representation and object detection performance, indicating potential for generalization beyond the original detection dataset. These results provide empirical support for its application in public safety surveillance and other object detection tasks.

The above results indicate that the performance improvements brought by ACG-YOLO primarily originate from the network architecture itself, rather than from the statistical characteristics of any specific dataset. This further confirms that the proposed collaborative optimization framework possesses strong cross-scene adaptability and promising engineering application potential.

4.8. Qualitative Analysis

To provide a more intuitive evaluation of the detection performance of the proposed method, we further present both the detection result visualizations under typical scenarios and an accuracy–efficiency trade-off analysis.

4.8.1. Detection Result Visualization

To visually demonstrate the detection performance of the ACG-YOLO model, we select typical scenarios from the Violence and Weapon Detection dataset for visualization analysis. Figure 6 presents a comparison of detection results between ACG-YOLO and the baseline model in representative violent scenarios, with the left column showing the results of ACG-YOLO and the right column those of the baseline model.

As shown in Figure 6a, the baseline model fails to detect the small weapon target due to its limited size and partial occlusion, whereas ACG-YOLO successfully identifies it. This example is consistent with the role of CoordAtt in enhancing spatially informed feature representation for small targets. In Figure 6b, which involves multiple interacting individuals, ACG-YOLO achieves more stable localization than the baseline. This result is consistent

with the intended role of GCEM in incorporating global contextual information into feature representation. As illustrated in Figure 6c, under low-light and noisy surveillance conditions, ACG-YOLO remains more robust and is less affected by background interference compared to the baseline. This demonstrates that the proposed collaborative framework maintains reliable detection performance in challenging real-world environments.

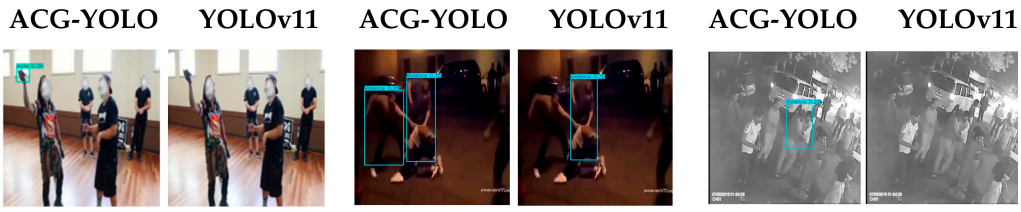


Figure 6. Qualitative comparison of detection results between ACG-YOLO and the baseline YOLOv11n. The blue boxes denote the predicted bounding boxes. (**a**) Small weapon target. (**b**) Complex violent interaction. (**c**) Low-quality surveillance scene.

Overall, the qualitative results show improved detection of small objects and more stable predictions in complex and low-quality surveillance scenes. These qualitative results are consistent with the quantitative experiments, further validating the effectiveness of the collaborative optimization among ADown, CoordAtt, and GCEM.

4.8.2. Accuracy–Efficiency Trade-Off Analysis

To more intuitively illustrate the trade-off among detection accuracy, computational efficiency, and model size for ACG-YOLO, Figure 7 presents a bubble chart of different algorithms on the Violence and Weapon Detection dataset. In this chart, the horizontal axis represents detection accuracy (mAP@0.5/%), the vertical axis denotes computational cost (GFLOPs), and the bubble size indicates the parameter count (Params/M), with smaller bubbles corresponding to more lightweight models.

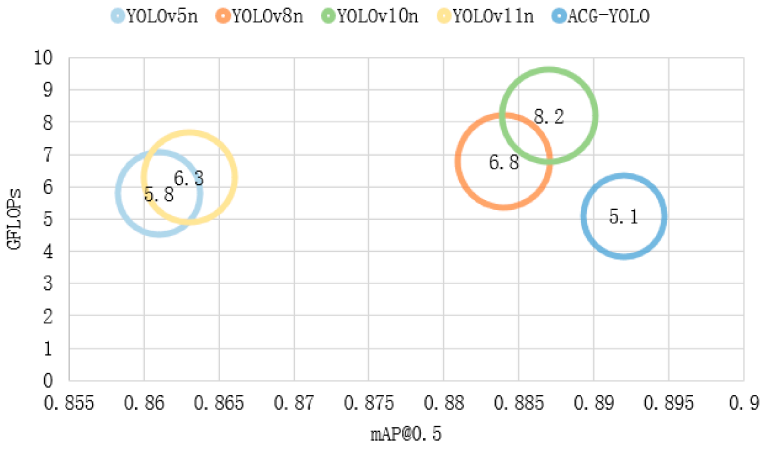


Figure 7. Precision–efficiency trade-off chart.

Figure 7 shows that there is a general trade-off between detection accuracy and computational complexity across different models. Although YOLOv8n and YOLOv10n achieve relatively high detection accuracy, they incur greater computational overhead. In contrast, YOLOv5n and YOLOv11n exhibit comparatively lower computational complexity, yet their detection accuracy still leaves room for improvement. In contrast, ACG-YOLO simultaneously achieves the highest detection accuracy (89.2%), the lowest computational

cost (5.1 GFLOPs), and the smallest model parameter scale, resulting in a favorable accuracy–efficiency trade-off.

By synthesizing the aforementioned experimental results, it can be seen that ACG-YOLO does not achieve performance improvement by increasing model complexity, but rather by further enhancing detection accuracy while maintaining a lightweight design, thereby achieving a synergistic optimization of detection performance and computational efficiency. This indicates that the proposed method is capable of balancing real-time performance and detection accuracy on resource-constrained edge devices, indicating potential for resource-constrained deployment.

5. Conclusions

To address the challenges of small-target missed detection, difficulties in understanding complex interaction scenarios, and deployment constraints on edge devices in public safety surveillance for violence and hazardous weapon detection, this paper proposes a lightweight collaborative detection framework, termed ACG-YOLO. The framework integrates ADown, CoordAtt, and GCEM into YOLOv11n to address computational efficiency, spatial localization, and global contextual representation, respectively, with the goal of improving detection performance and real-time deployability in complex surveillance scenarios.

Specifically, the ADown module reduces model computational complexity via an efficient downsampling strategy while preserving more fine-grained feature information. CoordAtt enhances the model's spatial perception of small-sized weapon targets through explicit positional encoding. GCEM enhances global contextual representation in complex violent scenes through lightweight channel recalibration. These three modules correspond respectively to the optimization needs at three levels—feature preservation, spatial localization, and global semantic modeling—and together constitute the lightweight collaborative optimization framework of ACG-YOLO.

Experimental results show that the proposed method improves detection accuracy while reducing model complexity on the Violence and Weapon Detection dataset. Compared to the baseline YOLOv11n, ACG-YOLO improves mAP@0.5 by 2.9 percentage points, while reducing parameter count and computational cost by 19.9% and 19.0%, respectively, thereby achieving simultaneous optimization of detection accuracy and computational efficiency. Furthermore, experiments on the COCO2017 dataset further validate the generality and cross-scene generalization capability of the proposed method. Detection result visualizations and accuracy–efficiency trade-off analysis indicate that ACG-YOLO shows potential for small-object detection and real-time inference in resource-constrained surveillance scenarios.

Although the proposed method achieves satisfactory detection performance, it still has certain limitations. For instance, under challenging scenarios involving extreme occlusion, ultra-long-range targets, and severe motion blur, the detection performance of the model still leaves room for further improvement. In addition, this work primarily focuses on object detection in single frames and has not yet fully exploited the temporal information available in video sequences. Future work will consider incorporating temporal modeling and behavior understanding methods to further enhance the model's capability in recognizing complex public safety events, as well as conducting systematic deployment and benchmarking on actual resource-constrained edge devices (e.g., NVIDIA Jetson series) to validate its practical inference performance. Beyond static surveillance scenarios, the proposed lightweight framework also shows promising potential for deployment on mobile and UAV-based monitoring platforms. In such applications, the visual perception module must operate in conjunction with robust vehicle dynamics and disturbance-rejection control

strategies to maintain reliable detection performance under environmental perturbations such as wind gusts, vibration, and rapid attitude changes. Future work will explore the integration of ACG-YOLO with UAV onboard processing units and evaluate its performance under real flight conditions, as well as investigate model quantization and TensorRT optimization for further inference acceleration on edge devices.

Author Contributions: Conceptualization, H.W. and X.W.; methodology, H.W. and X.W.; software, H.W. and X.W.; validation, H.W., X.W. and G.Z.; formal analysis, H.W.; investigation, X.W.; resources, G.Z.; data curation, X.W.; writing—original draft preparation, H.W. and X.W.; writing—review and editing, H.W., X.W., G.Z. and Y.L.; visualization, X.W.; supervision, Y.L. and G.Z.; project administration, Y.L.; funding acquisition, G.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the National Natural Science Foundation of China (NSFC), grant number 62462064; the Education Science Planning Project of Yunnan Province in 2024, grant number BC24019; and the Yunnan Provincial Department of Education Science Research Fund, grant number 2024J0105. The APC was funded by the National Natural Science Foundation of China (NSFC), grant number 62462064.

Institutional Review Board Statement: Not applicable. All images used in this study were obtained from publicly available datasets, and no personally identifiable information was involved.

Informed Consent Statement: Not applicable. This study did not involve human participants, and no personally identifiable information was used.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

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